### 1.2.4 Magnetic separation

This method is applicable to ferromagnetic ores and it is based on the difference in the magnetic properties of the ore and the impurities. For example tin stone can be separated from the wolframite impurities which is magnetic. Similarly, ores such as chromite, pyrolusite having magnetic property can be removed from the non magnetic siliceous impurities. The crushed ore is poured on to an electromagnetic separator consisting of a belt moving over two rollers of which one is magnetic. The magnetic part of the ore is attracted towards the magnet and falls as a heap close to the magnetic region while the nonmagnetic part falls away from it as shown in the figure 1.2.

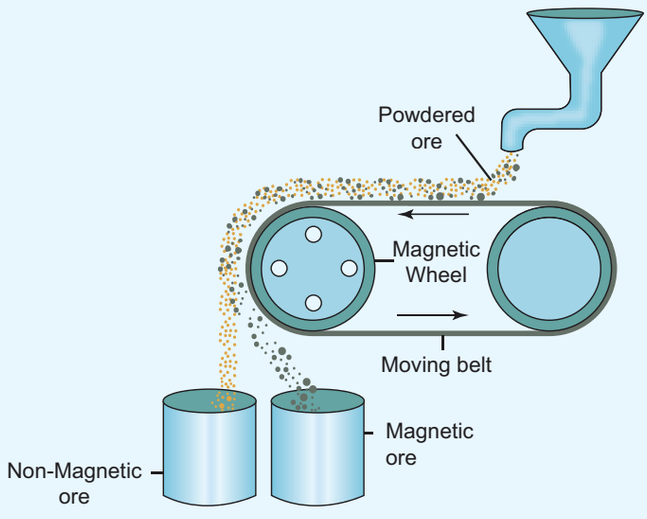



**Figure 1.2 Magnetic separation**

## 1.3 Extraction of crude metal

The extraction of crude metals from the concentrated ores is carried out in two steps namely, (i) conversion of the ore into oxides of the metal of interest and (ii) reduction of the metal oxides to elemental metals. In the concentrated ore, the metal exists in positive oxidation state and hence it is to be reduced to its elemental state. We can infer from the principles of thermodynamics, that the reduction of oxide is easier when compared to reduction of other compounds of metal and hence, before reduction, the ore is first converted into the oxide of metal of interest.

Let us discuss some of the common methods used to convert the concentrated ore into the oxides of the metal of interest.

### 1.3.1 Conversion of ores into oxides

#### Roasting

Roasting is the method, usually applied for the conversion of sulphide ores into their oxides. In this method, the concentrated ore is oxidised by heating it with excess of oxygen in a suitable furnace below the melting point of the metal.



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